

Kevin Lira

📍 Raleigh, NC 27605 ✉ kevin@kevinws.com 📞 984 382 3351 🔗 kevinws.com in kevinwsbr 🌐 kevinwsbr

PhD Student in Computer Science (NCSU) researching large language models for automated software vulnerability repair. Experienced software engineer with expertise in web performance optimization, usability, and secure code generation.

Education

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| North Carolina State University , Raleigh, NC
<i>PhD in Computer Science</i> | <i>Aug 2025 – Aug 2029
(Expected)</i> |
| ◦ Research on optimizing large language models for enhanced software security. | |
| Federal University of Alagoas , Maceió, AL, Brazil
<i>MS in Computer Science</i> | <i>2024 – 2026</i> |
| ◦ Thesis: Automated vulnerability repair using large language models. | |
| Federal University of Alagoas , Maceió, AL, Brazil
<i>BS in Computer Engineering</i> | <i>2017 – 2023</i> |
| ◦ Final paper: A large multi-language dataset of open-source software vulnerabilities and their fixes. | |

Experience

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| Research Assistant
<i>North Carolina State University</i> | <i>Raleigh, NC
Aug 2025 – Present</i> |
| ◦ Develop and evaluate LLM-based models for automated vulnerability repair. | |
| Engineering Technical Lead
<i>Atlas Technologies</i> | <i>Pelotas, RS, Brazil
Apr 2025 – Aug 2025</i> |
| ◦ Led UX enhancements that increased conversion rate by 0.32% and reduced bounce rate by 0.26%. | |
| ◦ Improved platform efficiency, SEO, and usability, raising funnel conversion to 18.6% and CSAT to 4.22/5. | |
| ◦ Led the design of a new content recommendation algorithm, boosting personalization and engagement. | |
| Front-end Engineer
<i>Atlas Technologies</i> | <i>Pelotas, RS, Brazil
Jan 2022 – Apr 2025</i> |
| ◦ Developed and maintained automated tests for frontend components. | |
| ◦ Led performance and SEO initiatives, cutting INP by 30% and reducing load times by 48%. | |
| ◦ Contributed to the internationalization of products, enabling multilingual support and global scalability. | |
| Front-end Developer
<i>ROGA</i> | <i>Maceió, AL, Brazil
Mar 2021 – Dec 2021</i> |
| ◦ Led legacy project migrations for modernization and performance. | |
| ◦ Migrated components to a shared library using Web Components for better reuse and modularity. | |

Publications

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- Vulnerability Detection with Interprocedural Context in Multiple Languages: Assessing Effectiveness and Cost of Modern LLMs**
30th International Conference on Evaluation and Assessment in Software Engineering (EASE 2026)
- Beyond Code Explanations: A Ray of Hope for Cross-Language Vulnerability Repair**
2nd ACM International Conference on AI-powered Software (AIware 2025)
- Turning Manual Tasks into Actions: Assessing the Effectiveness of Gemini-generated Selenium Tests**
2nd ACM International Conference on AI-powered Software (AIware 2025)

Technologies

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- Languages:** Python, TypeScript, C, SQL, JavaScript
- Frameworks & Libraries:** Node.js, Flask, Angular, Vue, Laravel
- Tools & Testing:** Git, Docker, Linux, Cypress, Playwright, Vitest